

FIRS Sampling Exam

Sampling Exercise

Planning for Zürichberg Post-Storm Remeasurement

The Zürichberg forest was inventoried the year before a major winter storm. The historical inventory contains measurements of stem number and basal area on field plots.

The forest owner now wants to know:

1. the total number of stems affected by the storm, and
2. the mean basal area affected by the storm.

Field remeasurement is expensive, so only a sample of the old plots can be remeasured. Your task is to propose **three different probability sampling designs**. For each design:

- explain how the sample will be selected and the reasoning for implementing it;
 - SRS without replacement
 - Two-stage sampling
 - cluster sampling
 - Stratification
 - etc...
- draw a realization of the sample to show to the owner;
- specify the estimator of total affected stems;
- specify the estimator of mean affected basal area;
- explain how the estimator will look and how uncertainty would be estimated;
- discuss advantages and disadvantages from the owner's perspective.

The owner doesn't know anything about sampling and will choose among your proposals so you will want to express yourself visually.

The dataset is loaded as follows:

```
library(forestinventory)
data(zberg)

zberg_forest <- zberg[!is.na(zberg$basal), ]
```

The package documentation describes **stem** as field-derived stem number and **basal** as field-derived basal area. It also provides **cluster**, **stade**, **melange**, and **couver** as potentially useful auxiliary information.

Estimation exercise